

PUNITH KM

+91 6361303701 | punithkmv@gmail.com | github.com/Punith-14 | linkedin.com/in/punith-km-91b831287

B.Tech CSE — Artificial Intelligence

PROFESSIONAL SUMMARY

AI-focused Computer Science student with practical experience architecting and deploying machine learning systems, NLP models, and deep learning architectures. Skilled in Python, Scikit-learn, XGBoost, and feature engineering across structured and unstructured data. Shipped a production-ready fraud detection system with 97.8% accuracy on 284,000+ transactions. Seeking a Machine Learning Engineer role to build, evaluate, and scale intelligent ML solutions.

EDUCATION

BTech — Computer Science Engineering (Artificial Intelligence)

Parul University | 2023 – Present | CGPA: 8.04

EXPERIENCE

Self-Employed | ML & AI Developer

2024 – Present

- Trained and evaluated ML models using Scikit-learn and XGBoost; measured precision, recall, and F1-score
- Used SMOTE oversampling and feature engineering on imbalanced datasets for classification tasks
- Developed NLP and computer vision models using deep learning frameworks, OpenCV, and MediaPipe
- Orchestrated multi-agent LLM workflows via LangChain and CrewAI for intelligent task automation
- Executed data preparation and EDA workflows across 3 project datasets totaling 290,000+ records

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, model training, evaluation (precision, recall, F1, ROC-AUC)

Deep Learning / NLP: Deep Learning, NLP, Computer Vision, OpenCV, MediaPipe

Data Science: Data Cleaning, EDA, Feature Engineering, SMOTE, Data Visualization

AI / LLM: LangChain, LangGraph, CrewAI, Prompt Engineering

Programming: Python, Java, C, JavaScript

Databases: MySQL, MongoDB

DevOps: Aws

Tools: Git, GitHub, Tableau, MS Excel, Streamlit

PROJECTS

Credit Card Fraud Detection & Analytics Pipeline | Python · Scikit-learn · XGBoost · SMOTE · Tableau | Completed

- Trained a fraud detection system on 284,000+ transactions achieving 97.8% accuracy and 91% recall
- Used SMOTE to address 0.17% class imbalance; extracted 10+ engineered features including transaction velocity and spending deviation
- Benchmarked 3 models — Logistic Regression, Random Forest, and XGBoost — on ROC-AUC and F1-score, cutting false positive rate by 23%
- Produced an interactive Tableau dashboard with fraud heatmaps, hourly anomaly trends, and risk scorecards

Gesture-Recognition Drone Coordination Network | Python · OpenCV · MediaPipe · Deep Learning | Completed

- Constructed a real-time computer vision model for gesture classification achieving sub-100ms inference latency
- Optimized failure detection and recovery logic across 5+ simultaneous nodes for resilient coordination
- Shipped a Streamlit monitoring dashboard for live system-status visualization of 10+ metrics

Tourist Guide Recommendation System | Python · NLP · HTML · CSS · JavaScript | Completed

- Formulated a recommendation engine covering 50+ tourist locations across 3 categories using location-aware logic
- Merged a Python ML backend with a JavaScript frontend, serving recommendations with under 2s response time

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Workshop — IIT Bombay Techfest (national-level technical festival)
- Data Science Workshop — Data Cleaning, Preprocessing & Feature Engineering
- Agentic AI Workshop — Advanced AI systems & multi-agent orchestration
- Prompt Engineering — Advanced techniques for large language models
- National-level SIH participant | Greenbottle sustainable product concept presented to industry judges

KEY STRENGTHS

Problem Solving | System Design | Analytical Thinking | Team Collaboration